

Fetal Movement Dataset Recorded Using Four Inertial Measurement Units

1 Introduction

Fetal movement monitoring is one of the most common standard method of monitoring fetal well-being. This is used by almost every pregnant mother around the globe to identify any deterioration of fetal well-being. With the hope of developing a non-invasive, low cost system to monitor fetal movements at home, following device was designed and fabricated. The device mentioned in Section 2 is then used to take readings from a single pregnant mother during her pregnancy.

2 Instrument

The instrument used for data acquisition consists of four Inertial Measurement Units (IMUs) (MPU6050), each of which contains a tri-axial accelerometer and a gyroscope, and a Raspberry Pi 1. For data retrieval, the Raspberry Pi reads each IMU sequentially in the order marked by numbers in Fig. 1. The sampling frequency is approximately 30Hz. Moreover, when taking measurements, the IMU units were mounted on the pregnant mother's abdomen as illustrated in Fig. 1; whenever a fetal movement occurs, the pregnant mother was advised to push the corresponding push button in the instrument. The instrument was designed under the guidance of a consultant obstetrician and gynaecologist.

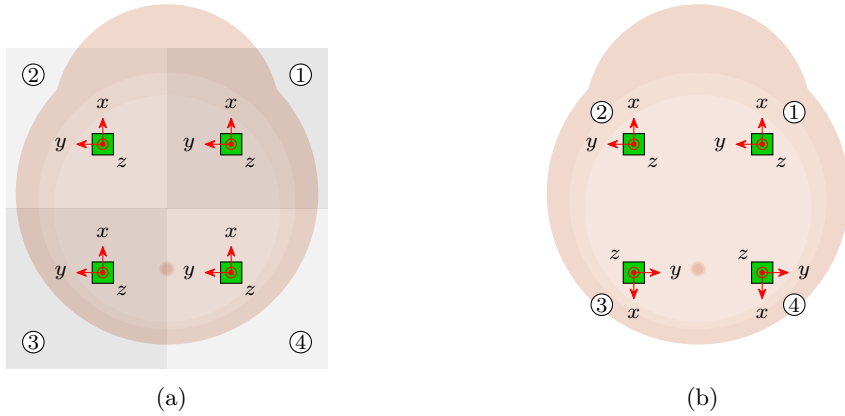


Figure 1: Illustration of the front view of pregnant mother's abdomen showing the sensor placement. Small green squares represent IMUs, with their axes marked. Two configurations are shown: (a) configuration used for sub-dataset one and two, where the gray squares superimposed on the top of the image mark the respective quadrants; (b) configuration used for sub-dataset three (note that the accelerometer three and four are rotated). Numbers denote the respective accelerometers and quadrants.

3 Data Format

The data file format is Comma Separated Value (CSV), where the top row contains the column names; an example data file is shown in Fig. 2. Some column names are described in Table 1, and the other column names follow the same format.

	time	ax1	ay1	az1	gx1	gy1	gz1	ax2	ay2	az2	...	gx3	gy3	gz3	ax4	ay4	az4	gx4	gy4	gz4	label
0	0.000000	12380	5448	8744	-216	-231	269	11356	-6592	9100	...	-946	306	-78	-15564	-2856	664	-468	133	37	0
1	0.033292	12484	5420	8816	-179	-219	267	11820	-6560	9060	...	-943	272	-105	-15420	-2860	584	-471	200	33	0
2	0.066273	12492	5424	8904	-161	-207	243	11836	-6516	8980	...	-972	300	-126	-15632	-2812	696	-483	231	63	0
3	0.099578	12480	5324	8720	-149	-510	224	11852	-6456	8872	...	-980	278	-167	-15644	-2788	752	-494	511	253	0
4	0.133043	12516	5372	8564	-168	-295	194	11784	-6412	8960	...	-958	266	-182	-15532	-2672	752	-263	203	252	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
95037	3262.171046	11576	7260	8684	38	-87	47	11980	-5372	9172	...	-1140	-25	-25	-14652	-6136	1252	-637	285	-275	0
95038	3262.207182	11656	7232	8912	91	-95	13	11972	-5280	9304	...	-1086	6	-164	-14748	-5924	1256	-595	228	-238	0
95039	3262.240674	11496	7304	8816	10	-7	19	11988	-5172	9156	...	-1017	42	-126	-14856	-5916	1116	-580	142	-169	0
95040	3262.273723	11608	7308	8704	-96	-71	84	11948	-5384	9060	...	-1010	40	-103	-14716	-6016	1164	-624	115	-148	0
95041	3262.307030	11476	7076	8564	-27	-67	131	12024	-5612	9076	...	-1086	75	-121	-14424	-6040	1072	-568	185	-224	0

Figure 2: Example data file.

Table 1: Column name descriptions.

Name	Description
time	Time in seconds since the start of the recording.
axi	Accelerometer reading in the x-axis of the i^{th} IMU.
gyj	Gyroscope reading in the y-axis of the j^{th} IMU.
label	Ground-truth label (Further described in following sections).

4 Data Collection

Data collection is done in a home rather than a clinical setting, and the dataset consists of measurements taken only from a single pregnant mother due to the prevailing COVID-19 pandemic. However, it was conducted under the guidance of a consultant obstetrician and gynaecologist. All the readings are taken while the mother is in the Fowler’s position (seated position). Additional information about the mother as well as the baby can be found in Section 5. This dataset does not contain the ground-truth labels acquired from any medical device but the ground-truth labels acquired through the mother’s perception. Hence, the time difference between the occurrence of a fetal movement and the corresponding label has some variation. Additionally, there may be mislabelling leading to imperfect ground-truth labels. Nevertheless, we believe that the dataset as a whole will be useful, even without ground-truth labels, for tasks such as unsupervised learning and data analysis.

5 Additional Information

This section includes some additional information about the mother and the baby, which could be useful in analysis. Table 2 contains details about mother’s blood pressure during the pregnancy, recorded at the clinics by medical professionals. Moreover, vaccination details of the first dose of the COVID-19 vaccine, which was administered during the pregnancy, are also included.

Mother's Data

- Age : 32 years
- Height : 159.5cm
- BMI : 23
- Baby Count : Second Baby
- Complications : None
- POA at Delivery : 39 weeks + 6 days

Mother's Vaccination Details

- Vaccine : Sinopharm
- POA : 38 weeks + 2 days
- Dose : First Dose

Baby's Data

- Birth : Normal
- Condition : Healthy
- Count : Singleton
- Gender : Female
- Weight at Birth : 3200g
- Head Circumference at Birth : 33cm
- Length at Birth : 54cm

Table 2: Mother's blood pressure record.

POA	7w 6d	13w 4d	23w 3d	29w 4d	35w 4d	38w 4d
Blood Pressure	90/70	120/80	110/80	110/80	110/70	110/80

6 Dataset (from 26th week to 39th week)

The dataset contains tri-axial accelerometer and tri-axial gyroscope readings from the four IMUs and labels. There are three sub-datasets, which have different ground-truth labelling configurations, included in this dataset. Please note that the labelling is subjective to the mother's perception. The dataset, as a whole, contains recordings spanning 14 weeks from 26th week to 39th week and in total, about 71 hours of recordings.

Additional data are provided in three additional 'csv' files, which contains the record number, the Period of Amenorrhoea (POA), start time, and end time of each recording.

6.1 Sub-dataset One (from 28th week to 31st week)

In this sub-dataset, only the occurrence of the particular type of fetal movement known as the fetal kick is considered for ground-truth labelling. These are strong jabs or startles felt by the mother most probably due to strong trunk movements of the fetus. Hence, this version only contains positive labels ('1', '2', '3' and '4') for kicks, and other fetal movements are labelled as negative ('0'). In particular, the respective quadrant in which the fetal kick occurred as per the mother's perspective is recorded, for example, if the fetal kick occurred in the first quadrant (see Fig. 1), it will be labelled as '1'. Sensor placement configuration depicted in Fig. 1a was used for the data collection of this sub-dataset.

This dataset contains data collected almost every day from 28th week to 31st week of the pregnancy and in total, about 12 hours of recordings. Table 3 provides descriptions of labels.

Table 3: Label descriptions for sub-dataset one.

Label	Description
0	No fetal movement or fetal movement except fetal kick.
1	Fetal kick in the first quadrant.
2	Fetal kick in the second quadrant.
3	Fetal kick in the third quadrant.
4	Fetal kick in the fourth quadrant.

6.2 Sub-dataset Two (from 26th week to 30th week)

Following a similar approach to the previous sub-dataset, this also records quadrant information and contains data collected almost every day from 26th week to 30th week and in total, about 19 hours of recordings. All types of fetal movement felt by the mother – including trunk movement, isolated limb movement, and general body movement – were considered for ground truth-labelling as fetal movements in this sub-dataset. Table 4 provides descriptions of the labels for this sub-dataset. Sensor placement configuration depicted in Fig. 1a was used for the data collection of this sub-dataset.

Table 4: Label descriptions for sub-dataset two.

Label	Description
0	No fetal movement.
1	Fetal movement in the first quadrant.
2	Fetal movement in the second quadrant.
3	Fetal movement in the third quadrant.
4	Fetal movement in the fourth quadrant.

6.3 Sub-dataset Three (from 32nd week to 39th week)

In this sub-dataset, quadrant localization information is discarded, and the emphasis was given to the classification of different types of fetal movements. Three types of fetal movements were considered.

1. **Trunk Movement:** These are felt by the mother as strong jab or a startle.
2. **Isolated limb movement:** These are felt as simple flutters or taps by the mother.
3. **General body movement:** These are strong movements felt by mother due to rolling or stretching of the fetus.

This dataset contains data collected almost every day from 32nd week to 39th week and in total, about 40 hours of recordings. Table 5 provides descriptions of the labels for this sub-dataset. Sensor placement configuration depicted in Fig. 1b was used for the data collection of this sub-dataset.

Table 5: Label descriptions for sub-dataset three.

Label	Description
0	No fetal movement.
1	Trunk Movement.
2	Isolated limb movement.
3	General body movement.

Acknowledgement

We would like to extend our sincere gratitude to the pregnant mother who volunteered for the data collection during the COVID-19 pandemic.